

problem 8

$$u = (x^2 + y^2 + z^2)^{\frac{1}{2}} \quad \text{--- (1)}$$

by partially differentiating with respect to x , we get

$$u_x = \frac{d}{dx} (x^2 + y^2 + z^2)^{\frac{1}{2}}$$

$$= \frac{1}{2} (x^2 + y^2 + z^2)^{\frac{1}{2} - 1} \{2x + 0 + 0\}$$

$$= \frac{x}{(x^2 + y^2 + z^2)^{\frac{1}{2}}}$$

$$\therefore \frac{u}{v} = \frac{v \frac{d}{dx}(u) - u \frac{d}{dx}(v)}{v^2}$$

$$\therefore u_{xx} = \frac{(x^2 + y^2 + z^2)^{\frac{1}{2}} \frac{d}{dx}(x) - x \cdot \frac{d}{dx} (x^2 + y^2 + z^2)^{\frac{1}{2}}}{\{(x^2 + y^2 + z^2)^{\frac{1}{2}}\}^2}$$

$$= \frac{(x^2 + y^2 + z^2)^{\frac{1}{2}} - x \cdot \frac{(x) \frac{1}{2} (x^2 + y^2 + z^2)^{\frac{1}{2} - 1} - (1) \frac{1}{2} x}{(x^2 + y^2 + z^2)^{\frac{1}{2}}}}{(x^2 + y^2 + z^2) \frac{v - u \frac{d}{dx}(v)}{v^2}}$$

$$= \frac{\frac{x^2 + y^2 + z^2 - x^2}{(x^2 + y^2 + z^2)^{\frac{1}{2}}}}{(x^2 + y^2 + z^2)}$$

$$= \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

similarly,

$$u_{yy} = \frac{z^2 + x^2}{(x^2 + y^2 + z^2)^{3/2}}$$

also we have u.r of z

$$u_{zz} = \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\therefore u_{xx} + u_{yy} + u_{zz} = \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{3/2}} + \frac{z^2 + x^2}{(x^2 + y^2 + z^2)^{3/2}} + \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{y^2 + z^2 + z^2 + x^2 + x^2 + y^2}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{2(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{2(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{2}{\sqrt{x^2 + y^2 + z^2}} \left[\because u = (x^2 + y^2 + z^2)^{1/2} \right]$$

[Proved]

Problem : 9

(1) given,

$$u = (ax + by)^2 - (x^2 + y^2) \quad \text{where } a^2 + b^2 = 2$$

by partially differentiating with respect to x, y we get,

$$\begin{aligned} u_x &= \frac{d}{dx} \{ (ax + by)^2 - (x^2 + y^2) \} \\ &= \frac{d}{dx} (ax + by)^2 - \frac{d}{dx} (x^2 + y^2) \\ &= 2(ax + by) \cdot a - (2x + 0) \\ &= 2a(ax + by) - 2x \\ &= 2a^2x + 2aby - 2x \\ &= 2x(a^2 - 1) + 2aby \\ \therefore u_{xx} &= \frac{d}{dx} \{ 2x(a^2 - 1) + 2aby \} \\ &= 2(a^2 - 1) + 0 \\ &= 2(a^2 - 1) \end{aligned}$$

$$\therefore u_y = \frac{d}{dy}$$

$$\text{Similarly, } u_{yy} = 2(b^2 - 1)$$

$$\therefore u_{xx} + u_{yy} = 2(a^2 - 1) + 2(b^2 - 1)$$

$$\begin{aligned} &= 2(a^2 - 1 + b^2 - 1) = 2(a^2 + b^2 - 2) \\ &= 2(2 - 2) = 0 \end{aligned}$$

(showned)

⑪ given, $u = 4(\alpha x + \beta y + \gamma z)^2 - (x^2 + y^2 + z^2)$ ——— ①

by partially differentiating eqn 1 with respect to x, y, z we get

$$\begin{aligned} u_x &= \frac{d}{dx} \{ 4(\alpha x + \beta y + \gamma z)^2 - (x^2 + y^2 + z^2) \} \\ &= 2 \cdot 4(\alpha x + \beta y + \gamma z) \cdot (\alpha + 0 + 0) - (2x + 0 + 0) \\ &= 8\alpha(\alpha x + \beta y + \gamma z) - 2x \\ &= 8\alpha^2 x + 8\alpha\beta y + 8\alpha\gamma z - 2x \\ &= 2x(4\alpha^2 - 1) + 8\alpha\beta y + 8\alpha\gamma z \end{aligned}$$

$$\begin{aligned} \therefore u_{xx} &= \frac{d}{dx} \{ 2x(4\alpha^2 - 1) + 8\alpha\beta y + 8\alpha\gamma z \} \\ &= 2(4\alpha^2 - 1) + 0 + 0 \\ &= 2(4\alpha^2 - 1) \end{aligned}$$

Similarly,

$$u_{yy} = 2(4\beta^2 - 1)$$

$$u_{zz} = 2(4\gamma^2 - 1)$$

$$\left. \begin{array}{l} \text{Given,} \\ \alpha^2 + \beta^2 + \gamma^2 = \pi \end{array} \right\}$$

$$\begin{aligned} \therefore u_{xx} + u_{yy} + u_{zz} &= 2(4\alpha^2 - 1) + 2(4\beta^2 - 1) + 2(4\gamma^2 - 1) \\ &= 8\alpha^2 - 2 + 8\beta^2 - 2 + 8\gamma^2 - 2 \\ &= 8(\alpha^2 + \beta^2 + \gamma^2) - 6 \\ &= 8\pi - 6 \end{aligned}$$

(Ans)